American Computer Science League

2025-2026 ● Contest 1: Short Problems Solutions ● Senior Division

1. Computer Number Systems $100_2 \bullet (42_8 + 375_8)$ $= 100_2 \bullet 437_8$ $= 100_2 * 100\ 011\ 111_2$ $= 10001111100_2$ $= 100\ 0111\ 1100_2$ $= 47C_{16}$	D. 47C
2. Computer Number Systems To find the next year after 2025_{10} that is a palindrome: 2025_{10} / 16 = 126 r 9, 126 / 16 = 7 r 14, 7 / 16 = 0 r 7 so read the remainders backwards to get $7E9_{16}$. The next year that is a palindrome is 7F7.	C. 7F7
3. Recursive Functions $f(7) = f(10) - 1 = 10 - 1 = 9$ $f(10) = f(9) + 2 = 8 + 2 = 10$ $f(9) = f(8) + 2 = 6 + 2 = 8$ $f(8) = 8 - 2 = 6$ Now substitute backwards. Therefore, $f(f(7)) = f(9) = 8$ $f(f(f(7))) = f(8) = 6$	A. 6
4. Recursive Functions For this square fractal-like diagram, there are 4 squares at the first level and only 3 at the other levels. Therefore, 4*3 = 12 squares, then 12*3 = 36 squares, then 36*3 squares = 108 squares, and finally 108*3 = 324 squares.	D. 324

5. What Does This Program Do?

```
letters = "MDCLXVI"
num = 1776
ans = ""
for x = 0 to 6
    val = int(num/arr(x))
    for y = 1 to val
        ans = ans + letters[x]
    next y
    num = num % arr(x)
next x
output ans
```

This program finds the Roman numeral for the number 1776 which is MDCCLXXVI using the 7 array values that are given to you. This problem is in honor of the 250th Anniversary of the United States of America being celebrated in 2026.

B. MDCCLXXVI

6. What Does This Program Do?

```
n = 0
for x = 1 to 4
    z = 10 - x
    a = str(1089 * x)
    s = ""
    for y = 0 to len(a)-1
        s = a[y] + s
    next y
    if s == str(1089 * z)
        n = n + 1
    end if
next x
output n
```

This program uses a property of 1089 that these are reversals of each other:
1089*1=1089 and 1089*9=9801,
1089*2=2178 and 1089*8=8712,
1089*3=3267 and 1089*7=7623,
1089*4=4356 and 1089*6=6534.
All 4 are reversals. There is also a pattern in the digits for each product, so 1089*5=5445.

A. 4